THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 12	Take the lowest powers of common primes
2	**(b)** 29	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)$
3	**(a)** -2	Eliminate y
4	**(d)** $\frac{9}{2}$	Sum $=-b/a$
5	**(d)** 40	First find $d=3$
6	**(c)** 4	$AE=k\times AD$
7	**(a)** 24	Find the hypotenuse
8	**(a)** $\frac{3}{4}$	Substitute the standard values
9	**(c)** 20 m	Add the height of the eyes: $2+18$; Height of building $=20$ m
10	**(a)** 30 cm	$r^2=OP^2-PT^2=34^2-16^2=900$; $r=30$ cm
11	**(d)** 56 cm²	Circle $=154$ cm^2; square $=98$ cm^2; Difference $=56$ cm^2
12	**(c)** 225.5 m²	Canvas = CSA of cylinder + CSA of cone; $=88+137.5=225.5$ m^2
13	**(c)** 32.4	$\sum f_i=50$ and $\sum f_ix_i=1620$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{1620}{50}=32.4$
14	**(d)** $\frac{1}{4}$	Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $=9$; $P=\frac{9}{36}=\frac{1}{4}$
15	**(b)** 24	Take the highest powers of all primes
16	**(a)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
17	**(b)** 5	Eliminate x
18	**(b)** 2	Product $=c/a$
19	**(d)** 24	$n=\frac{l-a}{d}+1$
20	**(a)** 5	Perimeter ratio = side ratio
21	**(a)** 8	Midpoint
22	**(d)** 3	$\frac{c}{q}+\frac{p}{q}$
23	**(b)** $4\sqrt{3}$ m	$h(\cot 30^\circ-\cot 60^\circ)=8$ gives $h=8\div(\frac{2\sqrt{3}}{3})$; Height $=4\sqrt{3}$ m (distances 12 m and 4 m)
24	**(b)** 92 cm	Perimeter $=PA+AO+OB+BP=2(16)+2(30)$; Perimeter $=92$ cm
25	**(b)** 672 cm²	Circle area $=\frac{22}{7}\times 28^2=2464$ cm^2; square $=3136$ cm^2; Left over $=672$ cm^2
26	**(c)** 396 cm³	$\frac{2}{3}\pi(R^3-r^3)=\frac{2}{3}\times\frac{22}{7}(6^3-3^3)$; Volume $=396$ cm^3
27	**(a)** 26	Median class 20–30: $l=20$, $cf=16$, $f=15$, $h=10$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=20+\frac{25-16}{15}\times 10=26$
28	**(a)** $\frac{7}{8}$	Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $=7$; $P=\frac{7}{8}$
29	**(d)** 13	HCF of 65 and 117
30	**(b)** 10	$2^2-7(2)+k=0$
31	**(a)** 2	All ratios equal
32	**(b)** 7	Roots are 2, 9
33	**(c)** 1665	AP of two-digit multiples
34	**(a)** 6	$h^2=pq$
35	**(b)** 4	$\frac{x_1+x_2+x_3}{3}$
36	**(c)** $\frac{1}{2}$	$\frac{c}{p}-\frac{q}{p}$
37	**(b)** 24 m	The points are on opposite sides, so the distances add; Distance $=24$ m

38	**(a)** 360 cm²	Area $= 2 \times \frac{1}{2} \times 40 \times 9$; Area $= 360$ cm^2
39	**(d)** 1034 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(27^2 - 20^2)$; Area of path $= 1034$ m^2
40	**(c)** 6835.5 cm³	Cube $= 9261$ cm^3; hemisphere of radius 10.5 cm $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; Volume $= 6835.5$ cm^3
41	**(d)** 77	$l = 70$, $f_1 = 16$, $f_0 = 2$, $f_2 = 10$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$
42	**(d)** $\frac{7}{15}$	Favourable numbers: 3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30; number $= 14$; $P = \frac{14}{30} = \frac{7}{15}$
43	**(c)** 27	LCM + remainder
44	**(b)** 3	$2^2 + 2k - 10 = 0$
45	**(c)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
46	**(d)** 20	$16 - 36 + k = 0$
47	**(c)** 623	Use S_n
48	**(c)** 36	Similar triangles
49	**(d)** 5	Distance from centre
50	**(b)** 2	Each ratio is 1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** $3\sqrt{3}$ m	Distance from the foot $= 3\cot 30^\circ = 3\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 3\sqrt{3}$ m
2	**(a)** 96 cm	Half-chord $= \sqrt{52^2 - 20^2} = 48$ cm; Chord $= 2 \times 48 = 96$ cm
3	**(a)** 77 cm²	$\frac{60}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 77$ cm^2
4	**(d)** 256 cm	$\frac{4}{3}\pi(12)^3 = \pi(3)^2 L$; $L = 256$ cm
5	**(c)** 6	$\frac{1750+35p}{44+p} = 39.2$ gives $(35 - 39.2)p = -25.2$; $p = 6$
6	**(d)** 6	$\frac{x}{3+x} = 2 \times \frac{3}{3+x}$ gives $x = 2 \times 3$; $x = 6$
7	**(c)** non-terminating repeating	$210 = 2 \times 3 \times 5 \times 7$
8	**(b)** -7	$p = -(\alpha + \beta) = -(1 + 6)$
9	**(d)** 9	Add the two equations
10	**(c)** two distinct real roots	$D = 4$
11	**(d)** 23	Middle term $= \frac{a+l}{2}$
12	**(c)** 100	Half-diagonals are legs
13	**(a)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
14	**(d)** $\frac{7}{3}$	Divide by $\cos\theta$
15	**(d)** $3 + 2\sqrt{3}$ m	Distance from the foot $= 3\cot 60^\circ = \sqrt{3}$ m; Original height of the tree $= 3 + 2\sqrt{3}$ m
16	**(c)** 5 cm	$AF = AB - r = 11 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(11 - r) + (60 - r) = 61$ gives $r = 5$ cm
17	**(d)** 90°	$\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 14}$; $\theta = 90^\circ$
18	**(a)** 1 cm	$\frac{4}{3}\pi(3)^3 = \pi(6)^2 x$; $x = 1$ cm
19	**(d)** 11	$91.5 = 90 + \frac{50-(36+x)}{20} \times 10$ gives $x = 11$; $x = 11$; then $y = 100 - 68 - x = 21$
20	**(c)** 2	Cross-multiplying: $2(3 + x) = 1(8 + x)$; $x = 2$
21	**(c)** rational	$\sqrt{3} \times \sqrt{3} = 3$
22	**(a)** 4	Sum of zeroes $= 7$, so other zero $= 7 - 3$
23	**(c)** 27	$(x + y)(x - y) = 9 \times 3$
24	**(d)** 8	Factorise
25	**(a)** 4	$a = a_p - (p - 1)d$
26	**(c)** 13	Right triangle with legs d and $a - b$
27	**(d)** 10	Area $= 0$
28	**(d)** 70°	$\cos(90^\circ - \theta) = \sin\theta$
29	**(b)** $5\sqrt{3}$ m	$h = 10\sin 60^\circ = 10 \times \frac{\sqrt{3}}{2}$; Height $= 5\sqrt{3}$ m
30	**(b)** 43 cm	$8x + 3 = 5x + 18$ gives $x = 5$; $QA = 8(5) + 3 = 43$ cm
31	**(d)** 686 cm²	Sector $= 1886.5$ cm^2, triangle $= \frac{1}{2} \times 49 \times 49 = 1200.5$ cm^2; Segment $= 1886.5 - 1200.5 = 686$ cm^2
32	**(d)** 1056 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 10 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1056$ cm^2
33	**(d)** 22	$35 = 30 + \frac{50-(24+x)}{10} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 57 - x = 22$
34	**(b)** $\frac{1}{4}$	Favourable outcomes: (2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6); number $= 9$; $P = \frac{9}{36} = \frac{1}{4}$
35	**(a)** 15	Take the lowest powers of common primes
36	**(a)** 25	$(\alpha + \beta)^2 - 2\alpha\beta = 7^2 - 2(12)$
37	**(c)** 4	Eliminate y

38	**(b)** $-\frac{7}{3}$	Sum $= -b/a$
39	**(d)** 52	First find $d = 6$
40	**(c)** 9	$AE = k \times AD$
41	**(b)** 30	Find the hypotenuse
42	**(d)** $\frac{7}{6}$	Substitute the standard values
43	**(b)** $7 + 7\sqrt{3}$ m	$p + 14 = p\tan 60°$ gives $p(\sqrt{3} - 1) = 14$; $p = \frac{14}{\sqrt{3}-1} = 7 + 7\sqrt{3}$ m
44	**(b)** $110°$	$\angle APB = 180° - 35° - 35°$; $\angle APB = 110°$
45	**(a)** 1089 cm²	Side $= 132 \div 4 = 33$ cm; Area $= 33^2 = 1089$ cm^2
46	**(a)** 20328 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 42 \times 70 + 2 \times \frac{22}{7} \times 42^2 = 20328$ cm^2
47	**(c)** 30	$\frac{1200}{40}$
48	**(b)** $\frac{5}{11}$	Favourable letters: E, U, A, I, O; number $= 5$; $P = \frac{5}{11}$
49	**(b)** 60	Take the highest powers of all primes
50	**(c)** $\frac{5}{4}$	$\frac{\alpha + \beta}{\alpha\beta}$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	Eliminate x
2	**(c)** -2	Product $= c/a$
3	**(c)** 31	$n = \frac{l-a}{d} + 1$
4	**(c)** 7	Perimeter ratio = side ratio
5	**(a)** 6	Midpoint
6	**(a)** 5	$\frac{c}{q} + \frac{p}{q}$
7	**(d)** one-third of the distance of P from the foot	$\cot 60^\circ : \cot 30^\circ = 1 : 3$
8	**(d)** 13 cm	$OP^2 = 144 + 25 = 169$
9	**(b)** 42 cm²	Quadrant $= 154$ cm^2; square $= 196$ cm^2; Remaining $= 42$ cm^2
10	**(d)** 192	$n = \frac{24\times12\times18}{3^3}$; $n = 192$
11	**(d)** 112	$\sum f_i = 50$ and $\sum f_i x_i = 5600$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5600}{50} = 112$
12	**(a)** $\frac{5}{12}$	Favourable outcomes: 15 pairs; number $= 15$; $P = \frac{15}{36} = \frac{5}{12}$
13	**(d)** 18	HCF of 72 and 126
14	**(b)** 15	$3^2 - 8(3) + k = 0$
15	**(d)** 3	All ratios equal
16	**(b)** 8	Roots are -3, 5
17	**(b)** 1188	AP of two-digit multiples
18	**(a)** 6	$h^2 = pq$
19	**(a)** 3	$\frac{x_1+x_2+x_3}{3}$
20	**(a)** $\frac{2}{3}$	$\frac{c}{p} - \frac{q}{p}$
21	**(d)** $5 + 8\sqrt{3}$ m	Add the height of the eyes: $5 + 8\sqrt{3}$; Height of building $= 5 + 8\sqrt{3}$ m
22	**(b)** 28 cm	$r^2 = OP^2 - PT^2 = 35^2 - 21^2 = 784$; $r = 28$ cm
23	**(a)** 224 cm²	Circle $= 616$ cm^2; square $= 392$ cm^2; Difference $= 224$ cm^2
24	**(a)** 357.5 m²	Canvas = CSA of cylinder + CSA of cone; $= 220 + 137.5 = 357.5$ m^2
25	**(c)** 38	Median class 30–40: $l = 30$, $cf = 21$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-21}{5} \times 10 = 38$
26	**(a)** $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, HTT; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
27	**(d)** 62	LCM + remainder
28	**(a)** -1	$3^2 + 3k - 6 = 0$
29	**(a)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
30	**(c)** 30	$25 - 55 + k = 0$
31	**(b)** 640	Use S_n
32	**(a)** 64	Similar triangles
33	**(d)** 13	Distance from centre
34	**(d)** 2	Each ratio is 1
35	**(c)** $6\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 12$ gives $h = 12 \div (\frac{2\sqrt{3}}{3})$; Height $= 6\sqrt{3}$ m (distances 18 m and 6 m)
36	**(a)** 98 cm	Perimeter $= PA + AO + OB + BP = 2(21) + 2(28)$; Perimeter $= 98$ cm
37	**(c)** 1050 cm²	Circle area $= \frac{22}{7} \times 35^2 = 3850$ cm^2; square $= 4900$ cm^2; Left over $= 1050$ cm^2
38	**(b)** 1056 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(8^3 - 2^3)$; Volume $= 1056$ cm^3
39	**(a)** 85	$l = 80$, $f_1 = 13$, $f_0 = 5$, $f_2 = 5$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{8}{16} \times 10 = 85$

40	**(b)** $\frac{27}{40}$	Favourable numbers: 27 outcomes; number $= 27$; $P = \frac{27}{40}$
41	**(c)** terminating	$312500 = 2^2 \times 5^7$
42	**(c)** -9	$p = -(\alpha + \beta) = -(2 + 7)$
43	**(b)** 8	Add the two equations
44	**(a)** two equal real roots	$D = 0$
45	**(a)** 28	Middle term $= \frac{a+l}{2}$
46	**(d)** 169	Half-diagonals are legs
47	**(c)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
48	**(d)** $\frac{17}{5}$	Divide by $\cos\theta$
49	**(d)** 48 m	The points are on opposite sides, so the distances add; Distance $= 48$ m
50	**(b)** 972 cm²	Area $= 2 \times \frac{1}{2} \times 36 \times 27$; Area $= 972$ cm^2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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